

IOT 2025/2026 - Homework

Exercise 3

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Data

$N=4$ (number of tags)

$L_2=4$ (arbitration period for $N=r=2$)

$L_3=\frac{51}{8}$ (arbitration period for $N=r=3$)

Solution

I have to compute the efficiency η for different values of the initial frame size ($r_1 \in \{1, 2, 3, 4, 5, 6\}$).

Given that $\eta(r_1) = \frac{N}{L_N(r_1)}$ and $L_N(r_1) = r_1 + \sum_{i=0}^{N-1} P_{N,r_1}(S=i) \cdot L_{N-i}$, the first things to do is to compute

all the arbitration periods not given in the problem: $L_1=1$,
 $L_4 = 4 + P_{4,4}(S=0)L_4 + P_{4,4}(S=1)L_3 + P_{4,4}(S=2)L_2 + P_{4,4}(S=3)L_1$.

To obtain the probabilities, I have used a simple python script that enumerates and evaluate all the r^N cases: this approach is computationally heavy yet it can be used given the small numbers involved in this exercise. I report here the script and its output.

```
#!/usr/bin/env python3

from collections import Counter
import itertools

def framed_aloha_counts_enumeration(N, r, i):
    # Boundary checks
    if i < 0 or i > r or i > N:
        return 0, r**N if r > 0 else 1
    if r == 0:
        return (1, 1) if N == 0 and i == 0 else (0, 1)

    match_count = 0
    total_configurations = 0

    # Enumerate all possible configurations
    for config in itertools.product(range(r), repeat=N):
        total_configurations += 1
        slot_counts = Counter(config)

        # Count successful slots (slots with exactly 1 tag)
        successful_slots = sum(
            1 for slot, count in slot_counts.items() if count == 1
        )
        if successful_slots == i:
            match_count += 1
```

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    return match_count, total_configurations

if __name__ == "__main__":
    N_tags = 4
    r_sizes = [1, 2, 3, 4, 5, 6]

    for r in r_sizes:
        print(f"--- Frame Size r = {r} (Total Configurations: {r*N_tags}) ---")

        for i in range(r + 1):
            good_cases, total_cases = framed_aloha_counts_enumeration(
                N_tags, r, i
            )
            print(f" P(S={i}) = {good_cases}/{total_cases}")
        print()

```

```

--- Frame Size r = 1 (Total Configurations: 1) ---
P(S=0) = 1/1
P(S=1) = 0/1

--- Frame Size r = 2 (Total Configurations: 16) ---
P(S=0) = 8/16
P(S=1) = 8/16
P(S=2) = 0/16

--- Frame Size r = 3 (Total Configurations: 81) ---
P(S=0) = 21/81
P(S=1) = 24/81
P(S=2) = 36/81
P(S=3) = 0/81

--- Frame Size r = 4 (Total Configurations: 256) ---
P(S=0) = 40/256
P(S=1) = 48/256
P(S=2) = 144/256
P(S=3) = 0/256
P(S=4) = 24/256

--- Frame Size r = 5 (Total Configurations: 625) ---
P(S=0) = 65/625
P(S=1) = 80/625
P(S=2) = 360/625
P(S=3) = 0/625
P(S=4) = 120/625
P(S=5) = 0/625

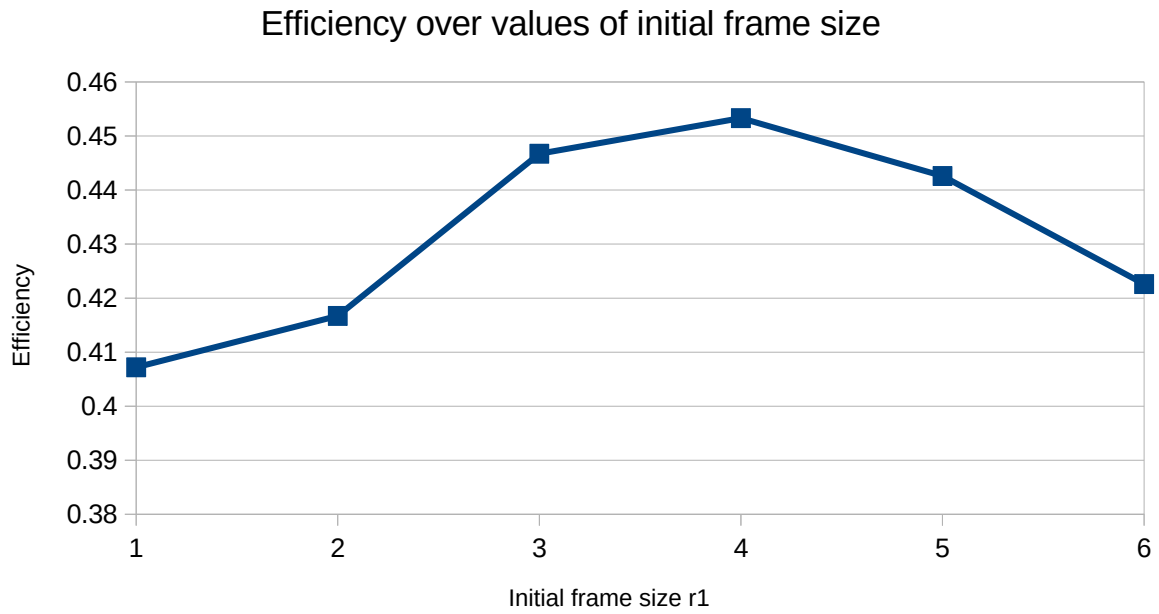
--- Frame Size r = 6 (Total Configurations: 1296) ---
P(S=0) = 96/1296
P(S=1) = 120/1296
P(S=2) = 720/1296
P(S=3) = 0/1296
P(S=4) = 360/1296
P(S=5) = 0/1296
P(S=6) = 0/1296

```

$$\text{So: } L_4 = \frac{4 + \frac{48}{256} \cdot \frac{51}{8} + \frac{144}{256} \cdot 4}{1 - \frac{40}{256}} = 8.824$$

Finally, I can compute the efficiency for each value of r_1 :

- $r_1=1$
 $L_4(1)=r_1+P_{4,1}(S=0)L_4+P_{4,1}(S=1)L_3+P_{4,1}(S=2)L_2+P_{4,1}(S=3)L_1=1+1\cdot 8.824=9.824$
 $\eta(1)=\frac{N}{L_4(1)}=0.4072=\mathbf{40.72\%}$
- $r_1=2$
 $L_4(2)=r_1+P_{4,2}(S=0)L_4+P_{4,2}(S=1)L_3+P_{4,2}(S=2)L_2+P_{4,2}(S=3)L_1=$
 $=2+\frac{1}{2}\cdot 8.824+\frac{1}{2}\cdot \frac{51}{8}=9.600$
 $\eta(2)=\frac{N}{L_4(2)}=0.4167=\mathbf{41.67\%}$
- $r_1=3$
 $L_4(3)=r_1+P_{4,3}(S=0)L_4+P_{4,3}(S=1)L_3+P_{4,3}(S=2)L_2+P_{4,3}(S=3)L_1=$
 $=3+\frac{21}{81}\cdot 8.824+\frac{24}{81}\cdot \frac{51}{8}+\frac{36}{81}\cdot 4=8.955$
 $\eta(3)=\frac{N}{L_4(3)}=0.4467=\mathbf{44.67\%}$
- $r_1=4$
 $L_4(4)=L_4=8.824$
 $\eta(4)=\frac{N}{L_4(4)}=0.4533=\mathbf{45.33\%}$
- $r_1=5$
 $L_4(5)=r_1+P_{4,5}(S=0)L_4+P_{4,5}(S=1)L_3+P_{4,5}(S=2)L_2+P_{4,5}(S=3)L_1=$
 $=5+\frac{65}{625}\cdot 8.824+\frac{80}{625}\cdot \frac{51}{8}+\frac{360}{625}\cdot 4=9.038$
 $\eta(5)=\frac{N}{L_4(5)}=0.4426=\mathbf{44.26\%}$
- $r_1=6$
 $L_4(6)=r_1+P_{4,6}(S=0)L_4+P_{4,6}(S=1)L_3+P_{4,6}(S=2)L_2+P_{4,6}(S=3)L_1=$
 $=6+\frac{96}{1296}\cdot 8.824+\frac{120}{1296}\cdot \frac{51}{8}+\frac{720}{1296}\cdot 4=9.466$
 $\eta(6)=\frac{N}{L_4(6)}=0.4226=\mathbf{42.26\%}$



The maximum value of the efficiency is reached when the number of slots in the initial frame is equal to the number of tags ($r_1=N=4$), according to the theory: this confirms that under or over allocating slots causes a degradation in efficiency.